

(KNN) Classifier Report

The classifier was evaluated using three different feature representations:

1. HOG features only
 2. CNN features only
 3. HOG + CNN features
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KNN Model Training

Hyperparameter Tuning

GridSearchCV was used to find the best KNN parameters.

Parameters Tested

- Number of neighbors (k):
 - 3
 - 5
 - 7
 - 9

The best k is 7

Distance Metric

- Euclidean distance

Weighting

- Distance weighting
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Feature Comparison Results

Three separate KNN models were trained and evaluated.

Feature Type	Validation Accuracy	Test Accuracy
HOG Only	0.580	0.6148
CNN Only	0.8673	0.8833
HOG + CNN	0.8853	0.8869

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HOG ONLY
Validation Accuracy: 0.5806451612903226
Test Accuracy: 0.6148409893992933

CNN ONLY
Validation Accuracy: 0.8673835125448028
Test Accuracy: 0.8833922261484098

HOG + CNN
Validation Accuracy: 0.8853046594982079
Test Accuracy: 0.8869257950530035
```

Analysis of Results

HOG Only

The HOG-only model achieved the lowest performance.

Reasons

- HOG captures only edges and gradients
- Sensitive to lighting and texture variation
- Cannot learn high-level semantic patterns

- Large feature dimension introduces noise for KNN

Although HOG was useful for shape representation, it was insufficient alone for accurate material recognition.

CNN Only

CNN features significantly improved performance.

Reasons

- ResNet50 learned rich visual representations
- Deep features capture texture and material appearance
- More discriminative than handcrafted features

CNN achieved high accuracy because pretrained deep features generalize well to image classification tasks.

HOG + CNN

The hybrid feature representation achieved the best overall performance.

Reasons

- HOG contributes local edge information
- CNN contributes semantic visual understanding
- Combining both creates richer feature representation

This combination improved:

- Overall accuracy
- Generalization
- Material discrimination

Therefore, the HOG + CNN model was selected as the final KNN classifier.

Final Selected Model

Best Model

- KNN using HOG + CNN features

Best Validation Accuracy

- 88.53%

Final Test Accuracy

- 88.69%

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===== BEST MODEL =====  
Best Feature Type : HOG+CNN  
Validation Accuracy : 0.8853  
Test Accuracy      : 0.8869
```

Classification Performance

Classification Report

Class	Precision	Recall	F1
Cardboard	0.97	0.95	0.96
Glass	0.85	0.86	0.85
Metal	0.86	0.90	0.88
Paper	0.94	0.96	0.95
Plastic	0.87	0.85	0.86
Trash	0.71	0.62	0.67

Classification Report (No Rejection):					
	precision	recall	f1-score	support	
0	0.97	0.95	0.96	38	
1	0.85	0.86	0.85	58	
2	0.86	0.90	0.88	48	
3	0.94	0.96	0.95	68	
4	0.87	0.85	0.86	55	
5	0.71	0.62	0.67	16	
accuracy			0.89	283	
macro avg	0.87	0.86	0.86	283	
weighted avg	0.89	0.89	0.89	283	

Discussion of Weakest Class

The Trash category achieved the lowest performance.

Reasons

- Small original dataset size
- Heavy augmentation required
- High visual variability
- Class imbalance before augmentation

Trash images are harder to classify because they often contain mixed materials and inconsistent visual patterns.

Rejection Mechanism

A rejection mechanism was implemented to improve prediction reliability.

Purpose

The rejection system prevents the classifier from making uncertain predictions.

Instead of forcing a prediction, low-confidence samples are labeled as:

- Unknown
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Rejection Threshold

A threshold of:

- 0.5

was used.

If confidence < 0.5:

- prediction becomes Unknown

Otherwise:

- prediction is accepted
-

Rejection Results

Metric	Value
Accuracy with rejection	90.07%
Rejected samples	11

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Accuracy with rejection: 0.9007352941176471
Rejected samples: 11
```

Conclusion

The KNN classifier successfully classified recyclable waste materials using image features extracted from HOG and ResNet50.

The experiments demonstrated that:

- Deep CNN features outperform handcrafted HOG features
- Hybrid HOG + CNN features produce the best results
- Confidence-based rejection improves reliability

The final KNN system achieved:

- 88.53% test accuracy
- 90.07% accuracy with rejection

This confirms that combining handcrafted and deep visual features can significantly improve material classification performance in waste management systems.